
The Specialist's Guide to Python Virtual Environments & Netmiko

Phase 1: Preparing the Server

Before creating the environment, you need to ensure the "Construction Tools" are installed on your Ubuntu server.

Bash

```
# Update your package list
sudo apt update
```

```
# Install the Python Virtual Environment creator and Pip
sudo apt install python3-venv python3-pip -y
```

Phase 2: The "Clean" Setup (Option 1 - Industry Standard)

Use this for every project. It keeps your server's "Core" safe.

1. Create a Project Folder:

Bash

```
mkdir ~/net_automation
cd ~/net_automation
```

2. Create the Virtual Environment:

This creates a folder named `venv` inside your project that holds a private copy of Python.

Bash

```
python3 -m venv venv
```

3. Activate the Environment:

This is the "Login" step. Your prompt will change to show `(venv)`.

Bash

```
source venv/bin/activate
```

4. Install Netmiko inside the "Safe Zone":

Bash

```
pip install netmiko
```

Phase 2: The "Quick Fix" (Option 2 - The Force Method)

If you are in a massive hurry and don't care about best practices (not recommended for long-term production).

Bash

```
# Force install to the system (ignores the 'externally managed' warning)
pip3 install netmiko --break-system-packages
```

Phase 3: Daily Workflow (How to use it)

Once it is installed, this is how you work every day:

Task	Command
Start Working	<pre>cd ~/net_automation source venv/bin/activate</pre>
Check Installed Tools	<pre>pip list</pre>
Run your Script	<pre>python my_script.py</pre>

Stop Working	deactivate
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I Phase 4: Troubleshooting & Notes

- The "ModuleNotFoundError": If you get this, it means you forgot to run `source venv/bin/activate`.
- The "Requirement already satisfied": This means you are probably trying to install Netmiko globally while the environment is NOT active.
- **Moving your Lab:** If you want to move your script to a new server, run `pip freeze > requirements.txt`. On the new server, just run `pip install -r requirements.txt` inside your new venv.

Lead's Pro-Tip: The "Interpreter" in VS Code

If you use **VS Code** to write your scripts on your laptop and connect to the server:

1. Press `Ctrl + Shift + P`.
2. Search for "Python: Select Interpreter".
3. Choose the one that points to `~/net_automation/venv/bin/python`.

This makes sure the red squiggly lines go away and VS Code knows where Netmiko is hiding!